CS532 Quiz 2: Epipolar Geometry

Name: _____

- **Q1:** Consider you have three images A,B,C, and you know a pair of correspondences in two images $x_A \leftrightarrow x_B$, but do not know the correspondece in the third image (C). How can knowing the Fundamental matrices relating the A-to-C (F_{AC}) and B-to-C (F_{BC}) help determine the coordinates of the correspondence in the third image?

  Solution:

  The fundamental matrix is a line generator. So we have:

  $$l_A^C = F_{AC} x_A$$
  $$l_B^C = F_{BC} x_B$$

  The corresponding point $x_C$lies at the intersection of these two lines: $\mathbf{x_C} = \mathbf{l_A^C} \times \mathbf{l_B^C}$

- **Q2:** Consider a pair of correspondences x and $x\prime$ and a known F matrix. If $x\prime = (1, 2, 1)^\intercal$ and $F = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}$, propose an epipolar consistent set of values for x.

  Solution:

  We have the epipolar constraint: $x\prime^\intercal F = x$.

  Using the values provided for $x\prime$ and F, we getthe line $x = (2, 4, 6)$.

  Therefore, any point that satisfies the line equation $\mathbf{x + 2y + 3 = 0}$ is epipolar consistent.